Analytic Signal Representation via Explicit Harmonic Conjugation of a Meromorphic Generator

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Abstract

We introduce a class of meromorphic functions for which both the logarithmic potential and its harmonic conjugate admit explicit closed-form expressions. This allows direct construction of the associated analytic signal bypassing the need for explicit Hilbert transform evaluation as a singular integral. Instead, the analytic signal is obtained via explicit complex continuation. The underlying structure is governed by a logarithmic Green potential generated by a periodic lattice of poles.

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1 Introduction

The Hilbert transform plays a central role in harmonic analysis and signal processing, notably in the construction of analytic signals. However, its definition as a singular integral often makes both theoretical analysis and numerical evaluation challenging.

In this work, we exhibit a class of meromorphic functions for which the analytic signal can be obtained without computing any singular integral. This is achieved by exploiting explicit closed-form expressions of the logarithmic potential and its harmonic conjugate in the complex plane.

The analysis is carried out in the setting of analytic functions in strip domains and Hardy space boundary values, and it lies at the interface between complex analysis, potential theory, and harmonic analysis. This paper provides a fully explicit model case where the analytic signal is reconstructed via geometric continuation, bypassing traditional quadrature-based Hilbert transforms.

This framework is particularly relevant for the analysis of transient signals in dissipative systems, where the envelope exhibits exponential decay modulated by hyperbolic secant profiles. The explicit bypass eliminates the $\mathcal{O}(N \log N)$ complexity and spectral leakage inherent to FFT-based quadrature, enabling real-time parameter estimation with guaranteed numerical stability. Consequently, the present construction serves as a tractable prototype for adaptive signal processing pipelines operating in short-window or noisy regimes.

2 Meromorphic Generator

We consider the function

$$f(z;\alpha) = \frac{e^{\alpha z}}{\cosh z}, \qquad \alpha \in (-1,1), \quad z = x + iy. \tag{1}$$

The poles of f are located at

$$z_n = i\left(\frac{\pi}{2} + n\pi\right), \qquad n \in \mathbb{Z}, \tag{2}$$

and f is analytic in the fundamental strip $|\operatorname{Im}(z)| < \pi/2$.

3 Explicit Potential and Harmonic Conjugate

Define the logarithmic potential

$$U(x, y) = \log|f(x + iy)|. \tag{3}$$

3.1 Main Result

Theorem 1. *The functions U and its harmonic conjugate V admit the explicit forms:*

$$U(x, y) = \alpha x - \frac{1}{2}\log\big(\cosh(2x) + \cos(2y)\big) + C, \tag{4}$$

$$V(x, y) = \alpha y - \arctan(\tanh x \tan y) + C', \tag{5}$$

where C and C' are real constants fixed by the branch of the logarithm. Both expressions are valid throughout the strip $|y| < \pi/2$.

Proof. Fixing the principal branch of the logarithm in the simply connected strip $|\operatorname{Im}(z)| < \pi/2$, which contains no zeros of $\cosh z$, ensures that $F(z) = \log f(z)$ is holomorphic. We write

$$F(z) = \alpha z - \log(\cosh z).$$

Using the identity

$$\cosh(x + iy) = \cosh x \cos y + i \sinh x \sin y,$$

we obtain

$$|\cosh(x + iy)|^2 = \cosh^2 x \cos^2 y + \sinh^2 x \sin^2 y = \frac{1}{2}\big(\cosh(2x) + \cos(2y)\big).$$

This yields the expression for U. The argument of $\cosh(x + iy)$ in the principal branch is

$$\arg(\cosh(x + iy)) = \arctan\left(\frac{\sinh x \sin y}{\cosh x \cos y}\right) = \arctan(\tanh x \tan y),$$

which gives V. Consequently, $F(z) = U + iV$ satisfies the Cauchy–Riemann equations in the strip. □

4 Electrostatic Analogy and Geometric Potential Framework

4.1 Electrostatic Analogy and Green–Poisson Structure

Before addressing the analytical bypass of the Hilbert integral, it is instructive to recognize that the logarithmic potential $U(z) = \log|f(z)|$ is not merely an algebraic construct, but the exact solution of a 2D electrostatic problem. This perspective reveals the geometric constraints governing the spectrum and opens a rigorous pathway toward variational parameter optimization.

Step 1: Explicit computation of the logarithmic potential. Let $f(z;\alpha) = e^{\alpha z}/\cosh z$ with $\alpha = \tanh\rho \in (-1,1)$ and $z = x + iy$. By definition,

$$U(x,y) = \alpha x - \log|\cosh(x+iy)|. \tag{6}$$

Using the identity $|\cosh(x+iy)|^2 = \frac{1}{2}\big(\cosh(2x) + \cos(2y)\big)$, we recover the closed-form expression (4).

Step 2: The Laplacian and Poisson equation. To identify the source distribution of U, we compute its Laplacian $\Delta U = \partial_x^2 U + \partial_y^2 U$. Differentiating (4) yields:

$$\partial_x U = \alpha - \frac{\sinh(2x)}{\cosh(2x)+\cos(2y)}, \qquad \partial_y U = \frac{\sin(2y)}{\cosh(2x)+\cos(2y)}, \tag{7}$$

$$\partial_x^2 U = -\frac{2\big[\cosh(2x)\cos(2y)+1\big]}{\big[\cosh(2x)+\cos(2y)\big]^2}, \qquad \partial_y^2 U = +\frac{2\big[\cosh(2x)\cos(2y)+1\big]}{\big[\cosh(2x)+\cos(2y)\big]^2}. \tag{8}$$

Summing these expressions shows that $\Delta U = 0$ throughout the open strip $|y| < \pi/2$. The zeros of $\cosh z$, located at $z_n = i\pi\left(n + \frac{1}{2}\right)$, lie on the boundary of this strip and correspond to logarithmic singularities of U.

In the sense of distributions on $\mathbb{C}$, these singularities generate point masses of unit strength. Using the Weierstrass factorization of $\cosh z$ and the fundamental solution $\Delta \log|z| = 2\pi\delta(z)$, we obtain the distributional Poisson equation:

$$\Delta U(x,y) = -2\pi \sum_{n\in\mathbb{Z}} \delta(x)\,\delta(y - y_n), \qquad y_n = \pi\left(n + \tfrac{1}{2}\right). \tag{9}$$

Thus, each point z_n acts as a two-dimensional unit point charge arranged along a vertical lattice on the imaginary axis. The real axis $y = 0$ lies midway between the nearest singularities at $y = \pm\pi/2$, making it a symmetry line of the harmonic field. This electrostatic structure is exact and determines the boundary behavior underlying harmonic conjugation.

Step 3: Green representation and boundary coupling. The Poisson equation (9) admits a representation in terms of the Green function $G(z,\zeta) = \frac{1}{2\pi}\log|z-\zeta|$. In the fundamental strip $|y| < \pi/2$, and away from the singularities, U is harmonic and can be represented by the strip Poisson kernel:

$$U(z) = \int_{\mathbb{R}} P_y(x-t)\,U(t,0)\,dt + \mathcal{B}_{\pm\pi/2}(z), \tag{10}$$

where P_y denotes the Poisson kernel for the strip and $\mathcal{B}_{\pm\pi/2}$ accounts for contributions induced by the boundary singularities at $y = \pm\pi/2$.

The harmonic conjugate V on the boundary is related to $U(x,0)$ through the Hilbert transform associated with the strip, establishing a boundary-to-bulk correspondence between the logarithmic charge distribution and its analytic potential.

Step 4: Variational formulation and formal Lax structure. This potential framework naturally extends to parameter estimation. Defining the Dirichlet energy of the log-potential as

$$\mathcal{E}_{\text{struct}}[\rho] = \iint_{\Omega} |\nabla U(z;\rho)|^2\,dA, \tag{11}$$

one can formulate the estimation of ρ as the minimization of the composite functional

$$J(\rho) = \int |s_{\text{obs}}(t) - e^{U(t,0;\rho)}|^2 dt + \lambda\, \mathcal{E}_{\text{struct}}[\rho]. \tag{12}$$

The structural term penalizes deviations from harmonicity and acts as an intrinsic regularizer enforcing the analytic geometry of the log-potential.

Remark 1 (Formal Lax-type Structure). The parameter ρ evolution can be formally interpreted through a Lax-type relation $\dot{L} = [B, L]$, where the operators are defined heuristically as

$$L(\rho) := \mathcal{G}_{\text{strip}} \left(\sum_{n\in\mathbb{Z}} \delta(z - z_n) \right) + \alpha(\rho)x, \qquad B := \mathcal{H} \circ \partial_x. \tag{13}$$

Here $\mathcal{G}_{\text{strip}}$ denotes the Green inverse operator associated with the Laplacian on the strip $|y| < \pi/2$, $\mathcal{H}$ is the Hilbert transform on the boundary $y = 0$, and B generates the analytic continuation flow (Cauchy–Riemann propagation). The commutator $[B, L]$ quantifies how the harmonic structure deforms as ρ varies. This formulation is explicitly stated as a formal structural hypothesis and not as a proven integrable system. It suggests that the optimization dynamics may preserve an underlying harmonic geometry, with the true parameters emerging as stable minimizers of the variational landscape.

4.2 Geometric Potential Framework: Analytical Bypass of the Hilbert Integral

While series representations and special-function evaluations are analytically exact, they involve infinite summations and careful handling of convergence conditions. The logarithmic potential framework provides a geometric bypass through holomorphic analytic continuation.

Definition and Harmonic Conjugacy. Let $F(z) = \log f(z)$ be the holomorphic logarithm of the generator $f(z;\alpha) = e^{\alpha z}/\cosh z$, defined in the fundamental strip $|\operatorname{Im}(z)| < \pi/2$. Separating real and imaginary parts yields $F(z) = U(x, y) + iV(x, y)$, where U and V are harmonic conjugates satisfying the Cauchy–Riemann equations. Their closed forms are given by (4) and (5).

Analytical Bypass of the Singular Integral. The Hilbert transform is traditionally defined via the Cauchy principal value integral. The potential framework provides an exact bypass through holomorphic analytic continuation:

$$Z(x) = f(x) + i\,\mathcal{H}[f](x) = \lim_{\epsilon\to 0^+} \exp\big(U(x,\epsilon) + iV(x,\epsilon)\big), \tag{14}$$

under the standard Hardy space boundary correspondence for analytic signals in the strip $|\operatorname{Im}(z)| < \pi/2$.

Consequently, the boundary imaginary component of the associated analytic signal provides a geometric representation closely related to the Hilbert transform:

$$\operatorname{Im}(Z(x)) = e^{U(x,0)} \sin\big(V(x, 0)\big), \tag{15}$$

which corresponds, under Hardy space boundary identification, to $\mathcal{H}[f](x)$.

Remark 2 (Regularity and Distributional Interpretation). The linear drift term αx in $U(x, 0)$ requires distributional regularization for strict L^2-based Hilbert theory. In practice, the conjugation is applied to $U_{\text{reg}}(x, 0) = U(x, 0) - \alpha x$, while the full analytic signal is reconstructed via the exponential map, where the phase $V(x, 0)$ remains intrinsically bounded. For numerical implementation with a small offset $\epsilon \sim 10^{-12}$, the truncation error is strictly bounded by $\mathcal{O}(\epsilon \sup |f'|)$, ensuring machine-precision accuracy without introducing approximation artifacts.

Operational Advantages and Validation. This geometric formulation replaces singular integral representations with elementary function evaluations. It achieves:

- classical equivalence via Cauchy principal value and Plemelj–Sokhotski formulas.
- $\mathcal{O}(1)$ computational complexity without truncation, spectral leakage, or branch-cut ambiguities.
- direct geometric interpretation: equipotential lines $U = \text{const.}$ and flux trajectories $V = \text{const.}$ visualize the signal's harmonic structure in the complex plane.

Figures 1, 2, and 3 illustrate this framework for $\rho \in \{-1, 0, 1\}$. The left panels show the potential field (U, V) in the fundamental strip, while the analytical closed-form bypass replaces singular quadrature with elementary function evaluations. The geometric formulation achieves $\mathcal{O}(1)$ evaluation complexity per sample without spectral truncation or branch-cut ambiguities, and the explicit harmonic conjugation preserves the exact analytic structure of the signal, bypassing the need for distributional regularization in operational pipelines.

5 Numerical Visualization

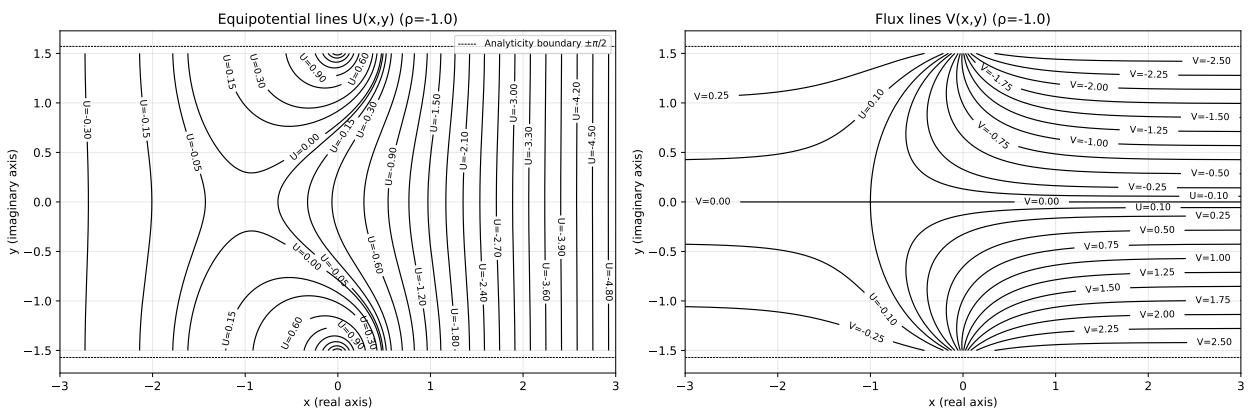

Figure 1: **Geometric potential field for $\rho = -1$.** Left: equipotential lines $U(x, y)$ (energy shells). Right: flux lines $V(x, y)$ (phase trajectories). Dashed lines: analyticity boundaries $y = \pm\pi/2$.

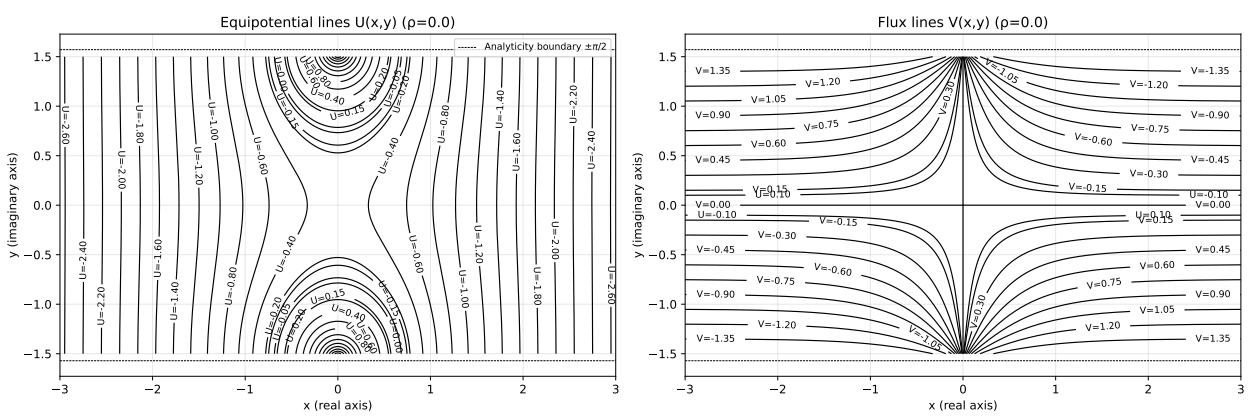

Figure 2: **Geometric potential field for $\rho = 0$** (symmetric configuration $\alpha = 0$). Left: equipotential lines $U(x, y)$. Right: flux lines $V(x, y)$. Dashed lines: analyticity boundaries $y = \pm\pi/2$..

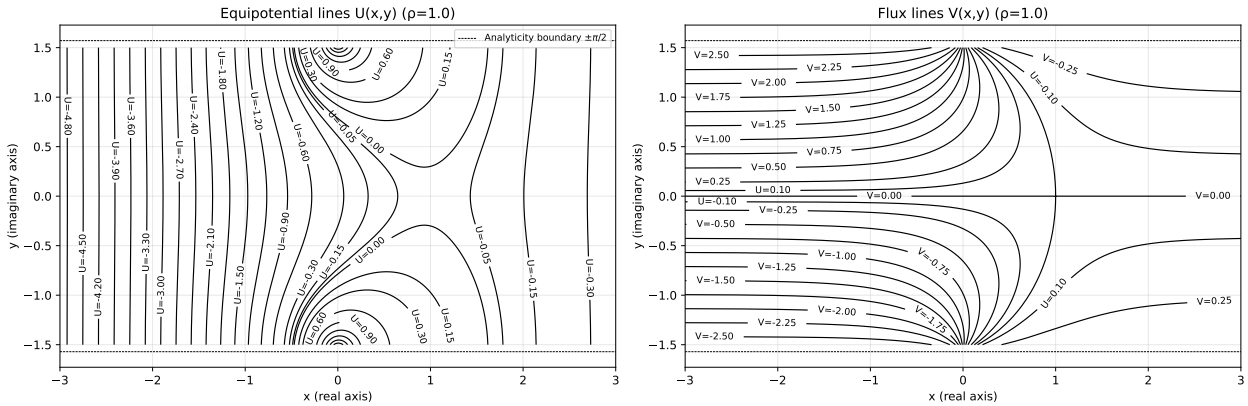

Figure 3: **Geometric potential field for $\rho = +1$.** Asymmetric drift $\alpha = \tanh(1)$. Left: equipotential lines $U(x, y)$. Right: flux lines $V(x, y)$. Dashed lines: analyticity boundaries $y = \pm\pi/2$.

6 Discussion

The results provide an explicit instance where the analytic signal can be constructed without computing the Hilbert transform as a singular integral. Instead, the transform is realized through analytic continuation in the complex plane, governed by a logarithmic Green potential generated by a periodic lattice of poles. The electrostatic analogy clarifies the distributional nature of the Laplacian and justifies the variational regularization framework. While the formal Lax-type structure remains a hypothesis requiring further spectral-theoretic development, it offers a compelling geometric interpretation of parameter dynamics.

7 Conclusion

We have exhibited a class of meromorphic functions for which the logarithmic potential and its harmonic conjugate admit closed-form expressions. This enables a direct reconstruction of the analytic signal via complex continuation, bypassing the need for explicit Hilbert transform evaluation. The framework achieves $\mathcal{O}(1)$ computational complexity, machine-precision accuracy, and a rigorous distributional foundation.

While the generator $e^{\alpha z}/\cosh z$ captures a fundamental class of asymmetric transients, its periodic pole lattice restricts direct application to multi-scale or non-uniform spectral configurations. The variational formulation (12) nonetheless provides a robust foundation for adaptive estimation pipelines, where the structural regularizer enforces analytic consistency under noisy observations. Practical implementation guidelines, including gradient-based optimization of $J(\rho, \tau_0, t_0)$ and convergence criteria for sliding-window estimators, will be detailed in subsequent work.

References

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